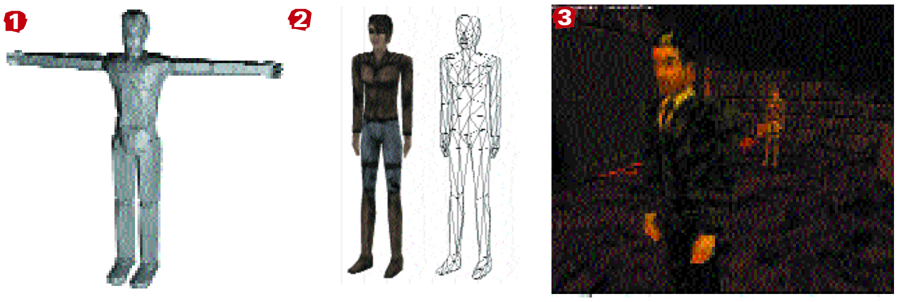

To star in a Machinima movie, a 3D model must first be created 1. It is then appropriately clothed 2 and finally animated 3

So is this real-time method all set to replace traditional computer animation? "No, I don't think it will," says Hancock. "I suspect that real-time production techniques will be some of the dominant film and CG techniques of this century, but just as theatre hasn't gone away with the advent of film, so I don't think that hand animation will disappear with this real-time revolution."

Frank Dellario, the Producer of the much-lauded *ILL Clan*, holds another view: "Yes and no. Yes, because the newest game engines and 3D cards are focusing not on increasing frame rates anymore but on the quality of the image. Also, once we have the ability to create and shoot Machinima in 48-bit colour vs 32-bit, we'll have the colour range to match the feature film world. No, because companies like Pixar Animation Studios will always use the newest, fastest hardware and software to push the envelope, like they did in *Monsters, Inc* by adding 3 million hair to the character Sulley. They didn't have to do that but they always will, they have to be the best and push what's possible. There were frames in *Luxo Jr* (Pixar's first animated short film, made in 1986) that took 72 hours to render. Today, *Luxo Jr* can be completely rendered in real-time. But *Monsters, Inc* also has frames that took just as long to render, almost 15 years later."

Real life problems

Stumbling blocks have prevented Machinima from gaining support amongst the community of traditional animators and filmmakers alike. Dellario feels that the technique needs to be understood, tried and tested by these groups. "With Machinima, an animation director will have to let go of key-frames and take on the role of a real-time director. He will have to sit in a room with puppeteers and direct them as actors on a live film set." But there are more practical concerns: "A major bottleneck is the licensing. Engine developers sell very expensive licensing agreements and share a percentage of profits from the games sold. Makes sense in the gaming industry; but for the Machinima community, they are simply out of league. Another obstacle is the toolsets. About every two years, newer game engines and computer hardware are released. It means having to develop whole new toolsets and/or having to recreate/upgrade our character models and world environments in a new format." Hancock shares Dellario's concern on licensing: "Probably the biggest hurdle at this point is a legal one, but I don't foresee engine licensing being a problem for long, as open-source alternatives are becoming increasingly powerful and viable."

In a way, Machinima is a blessed art form—it feeds off the fast-paced juggernaut that is the gaming industry. Games, as we all know, are at the zenith of technological advancements. If your computer can run the best game, it can pretty much run circles around any other application out there. The symbiotic relationship between game developers and computer hardware manufacturers has sparked off a revolution of late. Next-generation engines like those that power *DOOM III* and *Unreal Tournament (UT) 2003* are riding on the wave of advancement in video cards or programmable graphics processing units (GPUs) from the likes of ATi and nVidia, and promise to bring to shore an experience that will further blur the line between the digital and the real. Needless to say, the boys are excited at the prospect of these new toys. Hancock goes first: "*UT 2003* looks good, will have a lot of support behind it and generally seems like it could be great for Machinima. From what I've heard and seen, the 'Matinee' toolset built into the engine is a very solid offering, although it does seem to be very focused on the 'AI

Take a video card such as the Radeon 9700, add a next-generation game engine such as the one powering DOOM III and you can create movies like Final Fantasy: The Spirits Within on your PC, courtesy Machinima